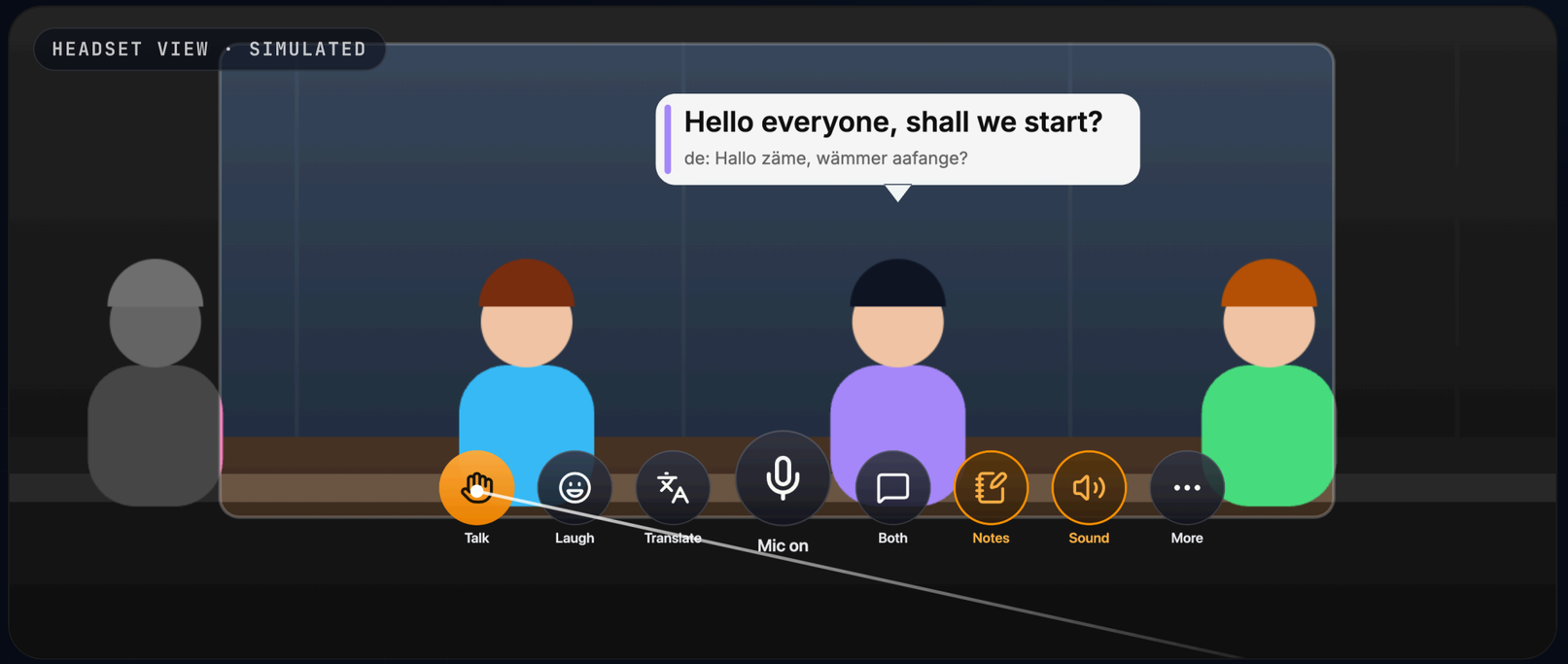
Progress: a meeting through a robot, end to end



- ✓ Head + body follow you, two-way audio
- ✓ Video pinned in the room, calibrated lens
- ✓ Looks at the speaker; talk + laugh
- ✓ Translated bubbles; Reachy speaks for you
- ◐ Room scan with depth (testing)



- live video, pinned in the room
- translated bubble
- dock: talk, laugh, translate, notes

2.1 s — was 7.1 s
you speak → Reachy says it, translated

20 /s — headset alone: ~3/s
faces tracked on the robot laptop

2.3 px — 24 views
lens calibration error, real camera

CORE PROBLEM · LATENCY

233 ms late — so we pin the picture in the room

VIDEO DELAY · DESKTOP BASELINE

233 ms

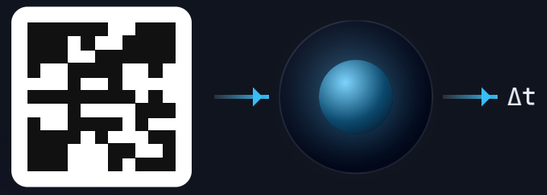
median, very steady

p10–p90: 217–234 ms · 300 frames

→ in the headset over Wi-Fi: measuring next

Pollen reports ~100 ms: Reachy Mini Wireless, hardware H.264, 5 GHz Wi-Fi.
Ours: Lite on a Mac, VP8 in software, 1080p.

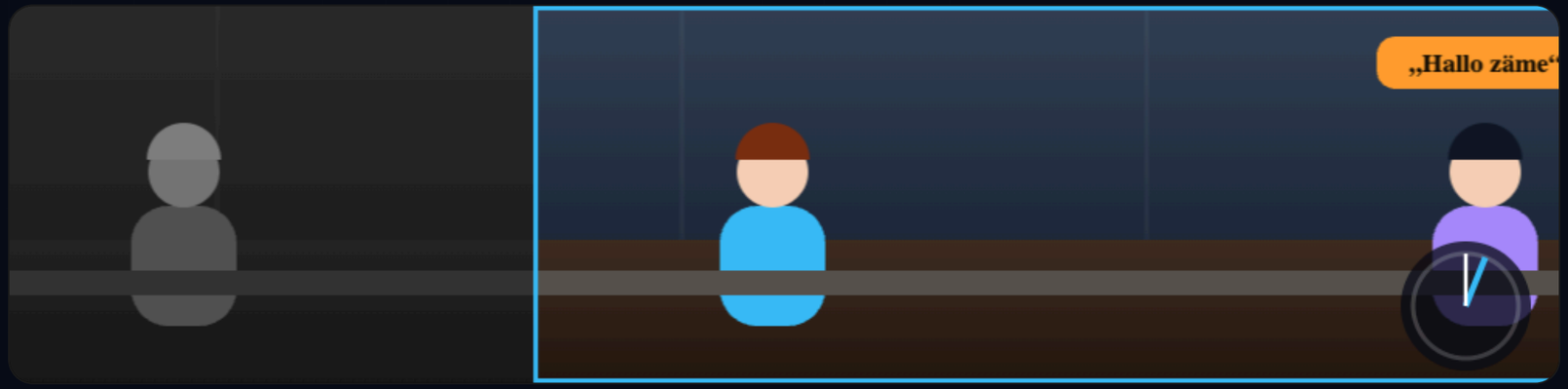
HOW WE MEASURE · QR LOOP



Naive: video glued to your head — 22° off — the room swims



Ours: pinned where the robot looked — 0° off — window catches up (22° behind)



room scan fills the edges — scan on | off

PROBLEMS & BLOCKERS

Two blockers, two lessons learned

BLOCKER 1 · FRAME RATE

Video costs half the VR frame rate

72 → 34 fps

Chrome copies every 1080p frame before VR can draw it.

→ frames straight to the GPU (built, testing) · plan B: Unity

BLOCKER 2 · DELAY

Our video is 2× later than Pollen's

233 vs ~100 ms

VP8 encoded in software, 1080p, on the laptop CPU.

→ test 720p and the Mac's hardware H.264 encoder



Robot laptop: Apple silicon only
An Intel Mac was too slow: its 908 ms headset run was invalid.

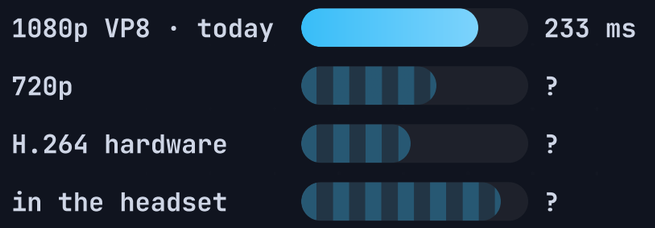


One Wi-Fi for headset and laptop
Otherwise the video connection never starts.

NEXT STEPS & TIMELINE

Six weeks to the live demo

① Latency study



→ Picture vs head error, reprojection on vs off.

② Unity prototype

WEB · JS

UNITY · C#

- pose.js
- camera.js
- speaker.js
- gestures.js

→ Faster video, face + eye tracking. Switch only if the numbers say so.

③ User study

P1 P2 P3 P4 P5

P6 P7 P8 P9 P10

→ 8–10 people, reprojection on vs off: time, sickness, presence.

Nov 2 · today
midterm

Nov 3–15
latency study

Nov 16–29
Unity prototype
go / no-go

Nov 30–Dec 11
user study,
demo rehearsal

Dec 14
final + live demo

Jan 11
report